FOR A MORPHISM OF FIBRED CATEGORIES THE SECTIONS OF THE LOCALISED COMPARISON MORPHISMS ARE COMPATIBLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the construction of the functor v' and its identification with the corresponding functor for p, and the transfer of the identities from the single-category lemma.

Editorial note. The label lemma-morphism-fibred-categories-with-object names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

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Allowed standard prerequisites (editorial).

- The section of the localised morphism for a single fibred category. Same chapter, the lemma constructing the functor v : D/p(U) -> C/U and the section sigma with pi_U o sigma = id and sigma^{-1} = pi_{U,*}. Supplies the construction applied twice, to p and to p'.
- Morphisms of fibred categories preserve strongly cartesian morphisms. Same chapter, the standing situation of a morphism of fibred categories over the base, whose defining property is the preservation of strongly cartesian morphisms. Supplies that u o v' is the functor attached to p and U.

Source reference. Original reference 08P9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9720–9720. Prerequisite pointer invoked at proof line(s) 9818; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08PA: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9767–9767. Prerequisite pointer invoked at proof line(s) 9797; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Lemma 0.1. *Assumptions and notation as in Situation 08PA. For $U' \in \mathrm{Ob}(\mathcal{C}')$ set $U = u(U')$ and $V = p'(U')$ and consider the induced morphisms of ringed topoi*

$$\begin{array}{ccc} (Sh(\mathcal{C}'/U'), \mathcal{O}_{U'}) & \xrightarrow{g'} & (Sh(\mathcal{C}), \mathcal{O}_U) \\ & {\scriptstyle \pi'_{U'}} \searrow \quad \swarrow {\scriptstyle \pi_U} & \\ & (Sh(\mathcal{D}/V), \mathcal{O}_V) & \end{array}$$

Then there exists a morphism of topoi

$$\sigma' : Sh(\mathcal{D}/V) \to Sh(\mathcal{C}'/U'),$$

such that setting $\sigma = g' \circ \sigma'$ we have $\pi'_{U'} \circ \sigma' = id$, $\pi_U \circ \sigma = id$, $(\sigma')^{-1} = \pi'_{U',*}$, and $\sigma^{-1} = \pi_{U,*}$.

Proof. Let $v' : \mathcal{D}/V \to \mathcal{C}'/U'$ be the functor constructed in the proof of Lemma 08P9 starting with $p' : \mathcal{C}' \to \mathcal{D}'$ and the object U'. Since u is a 1-morphism of fibred categories over $\mathcal{D}$ it transforms strongly cartesian morphisms into strongly cartesian morphisms, hence the functor $v = u \circ v'$ is the functor of the proof of Lemma 08P9 relative to $p : \mathcal{C} \to \mathcal{D}$ and U. Thus our lemma follows from that lemma. □